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1. Baking Soda, Baking Powder & Washing Soda

- Baking Soda (NaHCO_3 — Sodium Bicarbonate):
 - Preparation: Solvay process — $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$. [Q1458, Q1474, Q1506]
 - Heating: $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2\uparrow + \text{H}_2\text{O}$. [Q1462, Q1472]
 - Uses: Antacid (acidity relief), soda-acid fire extinguisher, baking, mouth freshening, deodorant, foot soak, hair care, bee sting relief (neutralizes formic acid), milk preservation (prevents acidisation), pesticide. [Q1448, Q1455, Q1461, Q1464, Q1471, Q1473, Q1494, Q1502]
 - NOT used for: Bleaching action. [Q1473]
- Baking Powder: Mixture of baking soda + mild edible acid (tartaric acid) + buffer (cornstarch). Dry chemical leavening agent. [Q1448, Q1459, Q1506]
- Washing Soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ — Sodium Carbonate Decahydrate):
 - Preparation: Recrystallisation of Na_2CO_3 — $\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$. [Q1450, Q1477]
 - Raw materials: NaCl , NH_3 , CO_2 (Solvay process). [Q1457]
 - Properties: White crystalline, basic salt, alkaline taste. [Q1469, Q1476]
 - Uses: Glass, soap, paper industries; borax manufacture; removing permanent hardness of water; domestic cleansing agent. [Q1450, Q1451, Q1467, Q1489]
 - NOT used for: Making cake (baking soda used). [Q1464]
- Chemical behaviour: Washing soda = basic salt (Q1469) / amphoteric nature bhi bataya gaya hai kuch explanations mein (Q1476). [Q1469, Q1476]

2. Bleaching Powder & Important Salts

- Bleaching Powder $\{\text{CaOCl}_2 / \text{Ca}(\text{OCl})_2$ — Calcium Hypochlorite}:
 - Preparation: Dry chlorine gas passed over dry slaked lime $\text{Ca}(\text{OH})_2$. [Q1453, Q1460, Q1468]

- Properties: Pale yellowish powder, strong chlorine smell, soluble in water (but never clear due to impurities), strong oxidizing agent. [Q1460]
- Uses: Bleaching cotton/linen/wood pulp, bleaching dirty clothes, oxidizing agent in industries, disinfectant, water decontamination. [Q1446, Q1453, Q1468]
- NOT used for: Baking cakes. [Q1446]
- Blue Vitriol (CuSO_4): Battery electrolytes. [Q1467]
- Green Vitriol (FeSO_4): Medicine for iron deficiency. [Q1467]
- Alum (Fitkari): Hydrated double salt — $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ (Potash alum); water purification mein use hota hai. [Q1383, Q1482]
- Ammonium Chloride (NH_4Cl): Nitrogen supply in fertilizers, electrolyte in dry cells. [Q1448]
- Whitewashing: Slaked lime $[\text{Ca}(\text{OH})_2]$ ki solution se walls whitewash karne ke 2-3 din baad, CO_2 (air se) reaction karke Calcium carbonate (CaCO_3) banata hai — yahi walls ko shiny finish deta hai. $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$. [Q1475]

3. Plaster of Paris, Gypsum & Cement

- Plaster of Paris (POP): $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ (Calcium Sulphate Hemihydrate).
 - Preparation: Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) ko 393 K par heat karke. [Q1352, Q1355, Q1369, Q1454]
 - Properties: White powder, water milane pe hard mass banata hai. [Q1454]
 - Uses: Setting fractured bones, dentistry, toys, decorative materials, cheap ornaments, cosmetics, blackboard chalk, fire-proofing material, sealing air-gaps in labs. [Q1445, Q1454]
 - NOT used for: Glass/soap/paper manufacture, domestic cleaning, baking. [Q1454]
- Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$): Soft dehydrated mineral; wallboard/ceiling manufacture; fluxing agent, fertilizer, paper filler, cement retarder. [Q1355, Q1465]
- Cement: Main compounds — Tricalcium silicate ($3\text{CaO} \cdot \text{SiO}_2$), Dicalcium silicate, Tricalcium aluminate, Tetracalcium aluminoferrite. CaO central component. [Q1470]

4. Caustic Soda, Alum & Other Compounds

- Caustic Soda (NaOH — Sodium Hydroxide):
 - Uses: Cleansing agent, washing soda manufacture, drain pipe cleaner, soap/detergent, paper & pulp industry, fabric industry. [Q1447]
 - NOT used in: Ammonia manufacture. [Q1447]
- Sodium Chloride (NaCl): Food preservative, healthcare (dehydration prevention). [Q1448]
- Milk of Magnesia $[\text{Mg}(\text{OH})_2]$: Heartburn, upset stomach, indigestion. [Q1489]
- Sodium Fluoride (NaF): Toothpaste. [Q1489]
- Titanium Dioxide (TiO_2): White pigment in paint. [Q1489]
- Sodium Nitrate (NaNO_3): Fertilizer. [Q1489]

5. Acids & Their Natural Sources

- Formic Acid (Methanoic Acid, CH_2O_2): Nettle leaves, ants, bee sting. [Q1452, Q1486]
- Citric Acid: Lemons, oranges, citrus fruits. [Q1452, Q1486]
- Tartaric Acid: Tamarind, apples, cherries, papaya, peach, pear, pineapple, strawberry, mango. [Q1452, Q1486]

- Oxalic Acid ($C_2H_2O_4$): Tomatoes. [Q1452]
- Lactic Acid: Sour milk, curd; produced during carbohydrate breakdown in body. [Q1486]
- Butyric Acid: Rancid butter ka foul smell; dairy products. [Q1487]
- Acetic Acid (Ethanoic Acid, CH_3COOH): Vinegar; pickled vegetables, sauces, raw material for spices. [Q1505]
- Stearic Acid: Animal & plant fats; soap preparation. [Q1487]
- Caproic Acid: Cheese-like smell. [Q1487]
- Palmitic Acid: Palm oil; cosmetics. [Q1487]

6. pH, Antacids & Toothpaste

- Toothpaste: Basic ($pH > 7$) — mouth mein bacteria se bane acid ko neutralize karta hai; pH 5.5 se neeche tooth decay hota hai. [Q1463]
- Antacids: Excess stomach acid neutralize karte hain.
 - Examples: $Al(OH)_3$, $MgCO_3$, $Mg_2O_8Si_3$, $Mg(OH)_2$, $CaCO_3$, $NaHCO_3$. [Q1353, Q1378, Q1466]
 - NaOH strong base hai — antacid ke roop mein use nahi hota. [Q1466]

7. Polymers & Plastics in Daily Life

- PVC (Polyvinyl Chloride): Water pipes, raincoats, handbags. [Q1483]
- Bakelite: Phenol + formaldehyde thermosetting polymer; electric switches, sockets, wire insulation, clocks, buttons, washing machines, toys, kitchenware. [Q1490, Q1492]
- Teflon (PTFE — Polytetrafluoroethylene): Non-stick coating on kitchen utensils. [Q1492]
- PET (Polyethylene Terephthalate): Bottles, utensils. [Q1488]
- Nylon: Backpacks, flak jackets, athletic shoes, umbrellas, swimsuits, ropes. [Q1488]
- Acrylic: Sweaters, fleece, socks, sportswear. [Q1488]
- Polystyrene: Food packaging, CD/DVD cases, building construction. [Q1483]
- Polyethylene: Plastic bags, bottles, films, containers, geomembranes. [Q1490, Q1492]
- Glyptal: Paints, metal coating, lacquers. [Q1483]

8. Gases & Their Everyday Uses

- Helium (He): 2nd lightest element (after H); colorless, odorless, tasteless; liquid at $-268.9^\circ C$; balloons (light + non-flammable); medicine, arc welding, refrigeration, aircraft gas, nuclear reactor coolant, cryogenic research. [Q1481]
- Nitrogen (N_2): Food preservation — displaces oxygen; atmosphere ka $\sim 78\%$; discovered by Daniel Rutherford (1772). [Q1501]
- LPG (Liquefied Petroleum Gas): Main component Butane (C_4H_{10}); also Propane (C_3H_8). Extracted from crude oil & natural gas. [Q1498]
- Biogas: 50–75% Methane (CH_4), 25–50% CO_2 , 2–8% N_2 ; produced from organic material in anaerobic environment. [Q1480]
- CO_2 : Bread/cake rise karata hai (baking soda se release); fire extinguisher. [Q1502]
- Electric bulbs: Filled with Nitrogen + Argon — tungsten filament oxidation rokne ke liye; bulb life badhata hai. [Q1493]
- Carbon Monoxide (CO): Odourless, colorless; incomplete combustion se; oxygen displace karke poisoning karta

hai. [Q1493]

9. Environmental Chemistry (Ozone, CFC, Biogas)

- Ozone (O_3): 3 oxygen atoms; stratosphere mein; highly reactive.
 - CFC (Chlorofluorocarbon) = ozone depletion ka main reason; man-made; used in refrigerants, aerosol sprays, foams, solvents. [Q1478, Q1479]
 - First ozone hole: 1970, Antarctica. [Q1478]
 - Montreal Protocol (1987): Global agreement to protect ozone layer. [Q1478]
 - Measurement unit: Dobson. [Q1478]
- Greenhouse gases: CO_2 , CH_4 , N_2O . [Q1480]

10. Metals, Coating & Corrosion Prevention

- Food cans: Tin se coated hain (NOT zinc) — kyunki zinc tin se zyada reactive hai; tin water/oxygen se unaffected rehta hai, sirf strong acids se react karta hai. [Q1484]
- Galvanization: Iron/steel pe zinc ki thin layer lagana — corrosion prevention. [Q1500]
- Anodizing: Protective oxide layer on metal (usually aluminum). [Q1500]
- Electroplating: Coating conductive surface with metal using electricity. [Q1500]

11. Physical Changes: Evaporation, Sublimation, Condensation

- Evaporation: Liquid $\rightarrow$ Gas; cooling effect.
 - Examples: Earthen pot mein water cool hona, acetone palm pe dalne se thandak, kapde sukhna, garam chai thandi hona. [Q1495, Q1496]
- Sublimation: Solid $\rightarrow$ Gas (no liquid).
 - Examples: Dry ice (solid CO_2), Iodine, Camphor, Naphthalene. [Q1496]
- Condensation: Gas $\rightarrow$ Liquid.
 - Examples: Morning dew, droplets on can/bottle, fog. [Q1496]
- Fusion: Two atoms combine to form heavier atom (e.g., $2H \rightarrow He$). [Q1496]
- Water: Freezing point $0^\circ C$ (solid + liquid coexist); boiling point $100^\circ C$; max density at $4^\circ C$; triple point $273.16 K$ ($0.01^\circ C$); absolute zero $0 K$ ($-273.15^\circ C$). [Q1497]

12. Separation Techniques

- Separating funnel: Immiscible liquids jaise oil & water ko alag karne ke liye. [Q1429, Q1499]
- Evaporation: Salt & water separate karne ke liye. [Q1499]
- Sedimentation: Heavier component settles when water added. [Q1499]
- Decantation: Water (with dust) removed from top. [Q1499]

13. Alcohols, Esters & Food Chemistry

- Ethanol (C_2H_5OH):
 - Uses: Cough syrups, alcoholic drinks, paint manufacturing, hand sanitizers, medical wipes (antibacterial/antifungal), motor fuel (gasohol), tincture iodine, tonics. [Q1485, Q1503]
 - Denatured alcohol: Methanol, benzene, pyridine, castor oil, gasoline, acetone milake — unsuitable for drinking; industrial solvent/fuel. [Q1395, Q1504]
- Methanol (CH_3OH): Wood alcohol; colorless, sweet pungent odor; alternative fuel; antidote for methanol poisoning = ethanol. [Q1491, Q1503]
- Acetic Acid (CH_3COOH) = Ethanoic acid: Vinegar; colorless, hygroscopic, pungent smell, sour taste; pickled vegetables, sauces. [Q1505]
- Esters: Fruit aroma; formed by esterification (carboxylic acid + alcohol + conc. H_2SO_4).
 - Example: Ethyl butyrate = pineapple smell. [Q1491]
- Rancidity: Fats/oils ka oxidation/hydrolysis; unpleasant smell; antioxidants/airtight containers/ N_2 flush se roka jaata hai. [Q1390, Q1487]
- Brahma Kamal (Himalayan flower): Antiseptic properties — fevers, bone ache, intestinal ailments, cough, cold. [Q1449]

14. Photography & Silver Compounds

- Black & white photography: Silver chloride ($AgCl$) + Silver bromide ($AgBr$) dono use hote hain.
 - $AgCl$: Disinfectant, antiseptic, mercury poisoning treatment. [Q1444]
 - $AgBr$: Emulsion component for photographic image development. [Q1444]
 - Silver oxide (Ag_2O): Mild oxidizing agent — aldehydes $\rightarrow$ carboxylic acids. [Q1444]

15. Graphite & Carbon Allotropes

- Graphite: Giant covalent structure; each carbon bonded to 3 others in hexagonal layers; good conductor of electricity.
 - Uses: Pencils, lubricants, crucibles, foundry facings, polishes, arc lamps, batteries, electric motor brushes, nuclear reactor cores. [Q1456]
- ⚠ Trap: Graphite bad conductor nahi hai — ye statement galat hai. [Q1456]

16. ⚠ Traps / Key Differences Summary

- ⚠ Trap: Baking soda formula $NaHCO_3$ hai — $CaSO_4 \cdot \frac{1}{2}H_2O$ POP ka formula hai. [Q1455]
- ⚠ Trap: POP ka use toys & decoration ke liye hai — glass/soap/paper ya baking ke liye nahi. [Q1454]
- ⚠ Trap: Bleaching powder ka use baking cakes ke liye nahi hai. [Q1446]
- ⚠ Trap: Caustic soda ($NaOH$) ka use ammonia manufacture mein nahi hota. [Q1447]
- ⚠ Trap: Baking soda ka use bleaching action ke liye nahi hota. [Q1473]
- ⚠ Trap: Washing soda = basic salt (Q1469) — amphoteric/acidic/neutral nahi; par kuch explanations mein amphoteric mention hai (Q1476). [Q1469, Q1476]
- ⚠ Trap: Graphite good conductor hai — bad conductor nahi. [Q1456]
- ⚠ Trap: Denatured alcohol = unsuitable for drinking; peene ke liye nahi banaaya jaata. [Q1504]
- ⚠ Trap: Food cans tin se coated hain — zinc se nahi (kyunki Zn zyada reactive hai). [Q1484]

- ⚠ Trap: LPG main component = Butane — methane/propane/ethane nahi. [Q1498]
- ⚠ Trap: Water 0°C pe solid + liquid donon state mein exist kar sakta hai. [Q1497]
- ⚠ Trap: Nitrogen food preservation ke liye use hoti hai — oxygen/hydrogen/chlorine nahi. [Q1501]
- ⚠ Trap: CO₂ bread/cake rise karata hai — H₂/CO/O₂ nahi. [Q1502]
- ⚠ Trap: Baking powder = Baking soda + Tartaric acid — calcium carbonate nahi. [Q1506]
- ⚠ Trap: Ethanol = cough syrups + alcoholic drinks + paints sab mein use hota hai — "All of the above". [Q1485]
- ⚠ Trap: Vinegar = Acetic acid — ethanoic acid ka hi dusra naam hai, par question mein specifically "acetic acid" correct tha. [Q1505]
- ⚠ Trap: Whitewashed walls ka shiny finish 2-3 din baad Calcium carbonate (CaCO₃) banne se aata hai — CO₂ air se reaction karta hai slaked lime ke saath. [Q1475]